Design of Full Subtractor

Group Number:- 24

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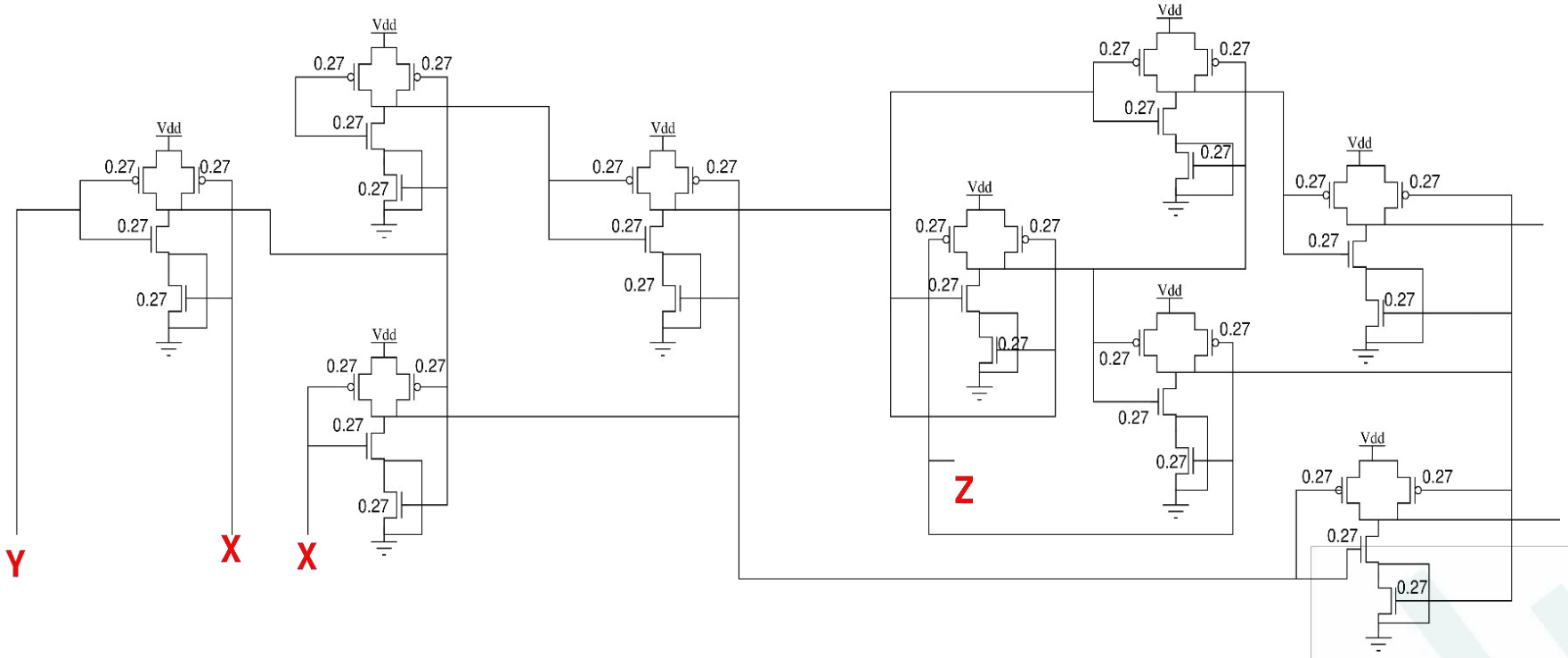
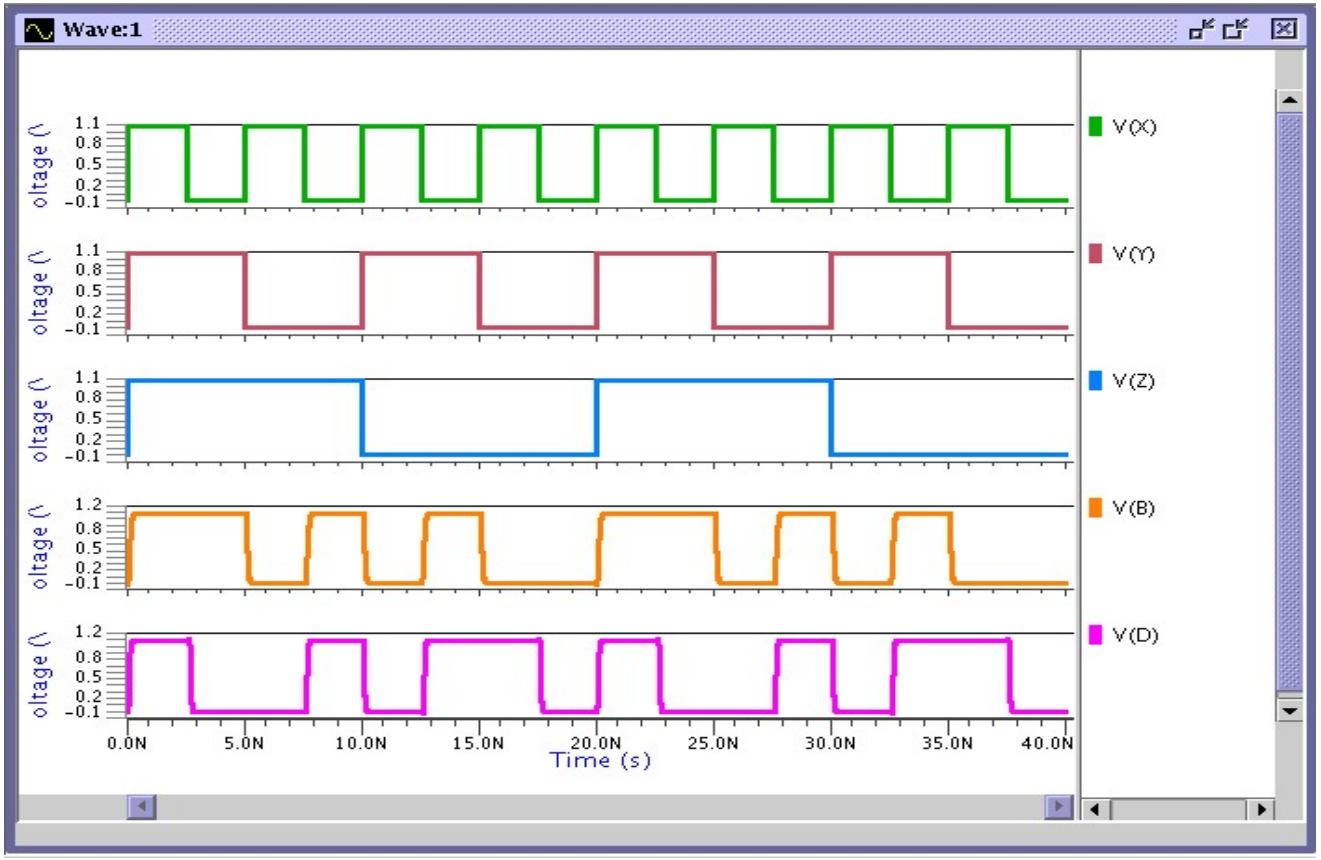
Schematic & Sizing



Functionality of the Circuit



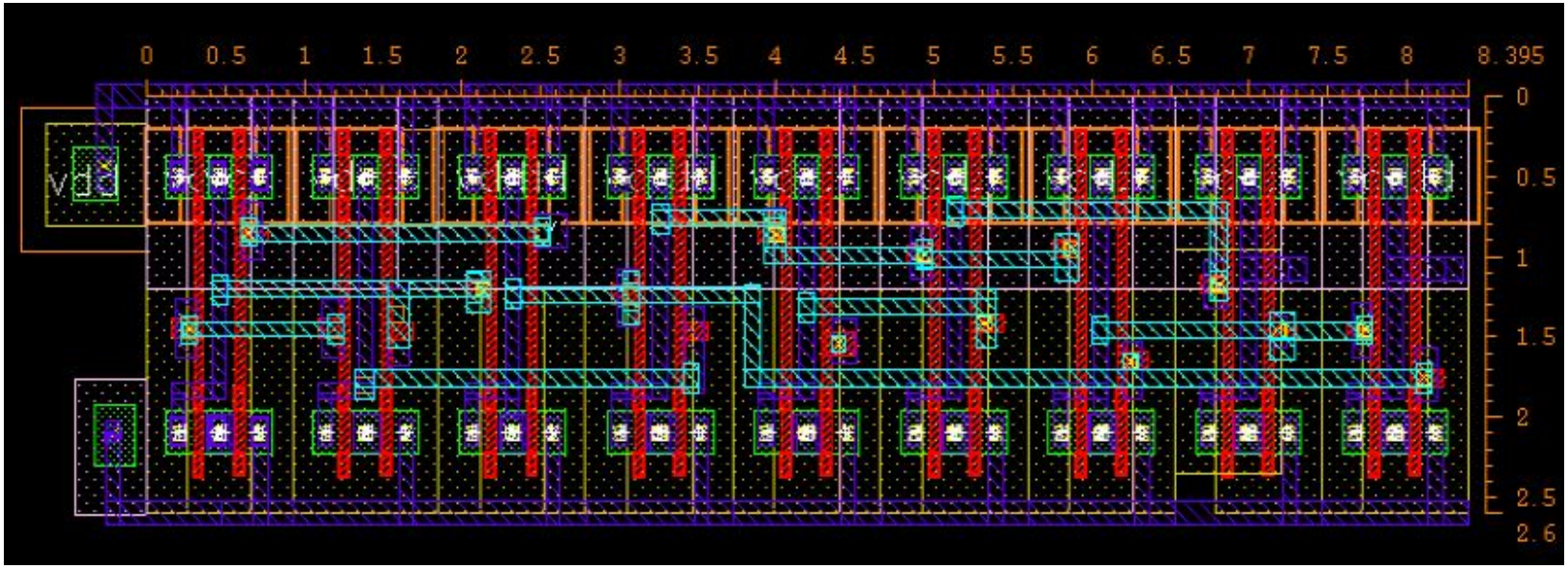
Truth Table for Full Subtractor

A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Input Stimuli

	DIFF	BORROW
TCD	001	111
TPD	101	110

Layout : Version 1

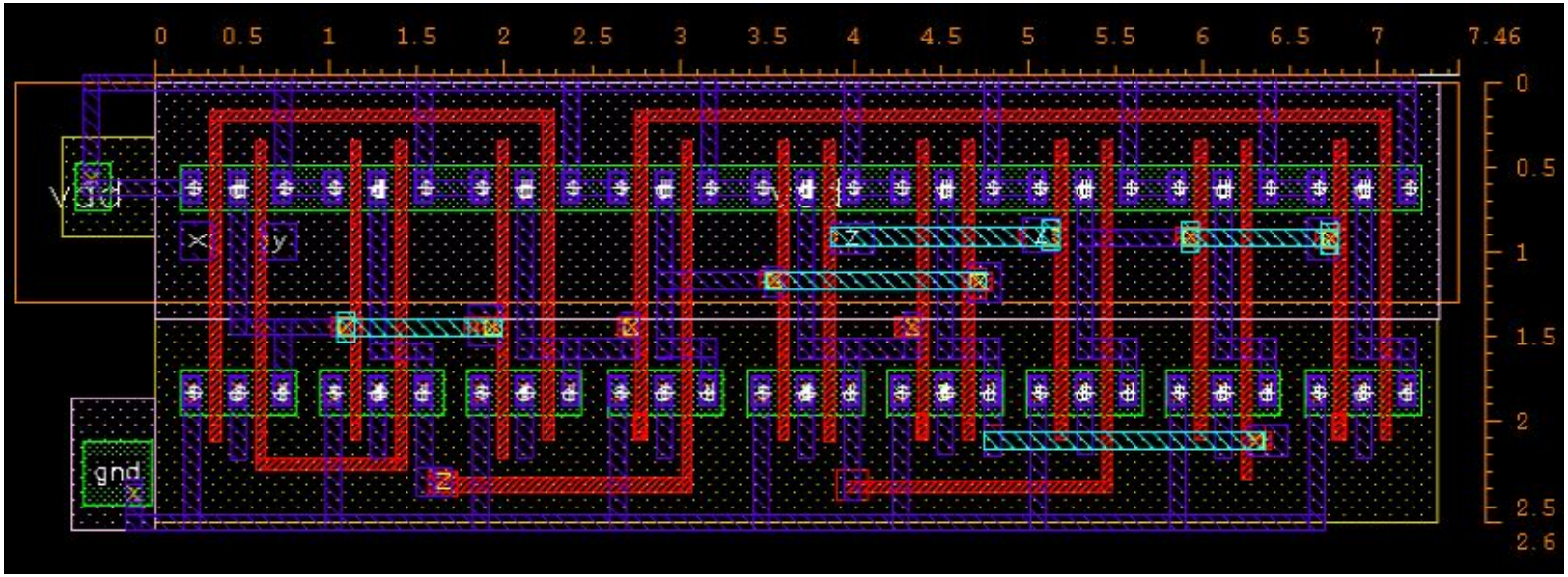


Version 1: Unoptimized Layout Design

1. Complex routing
2. More number of Metal2 Layer is used
3. Cell Height: 13 Tracks, Width: 42 Tracks
4. Larger Area is taken (21.83 μm^2)

This Layout can be optimized using proper floorplanning and routing.

Layout : Version 2



Version 2 : Optimized Layout Design

1. Planned Routing
2. Least number of Metal2 Layer is used
3. Cell Height: 13 Tracks, Width: 37 Tracks
4. Minimum Area is used (19.3 μm^2)
5. Less complexity offered

Shared Nwell and Diffusion, along with routing using poly wherever possible, result in lesser area and cost.

DRC/LVS Reports

Calibre - RVE v2013.1_34.21 : subtractor.drc.results

File View Highlight Tools Window Setup Help

Search

Show Unresolved subtractor, 0 Results (in 0 of 1579 Checks)

Check / Cell | Results

subtractor.cir (~/Desktop/eldo/subtractor/Post_Layout) - gedit

Calibre - RVE v2013.1_34.21 : svdb subtractor

File View Highlight Tools Window Setup Help

Search

Navigator

Results
- Extraction Results
- Comparison Results

ERC
- ERC Results
- ERC Summary

User Files
- LAYOUT_PDSP

Reports
- Extraction Report
- LVS Report
- Separate Properties

Rules
- Rules File

View
- Info
- Finder
- Schematics

Setup
- Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
subtractor	subtractor	23L, 23S	36L, 36S	7L, 7S

Cell subtractor Summary (Clean)

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                CELL COMPARISON RESULTS ( TOP LEVEL )

                     #      ##################
                    #       #                #      _   _
                  #   #     #    CORRECT     #       |
                   # #      #                #      \___/
                    #       ##################


LAYOUT CELL NAME:        subtractor
SOURCE CELL NAME:        subtractor

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NUMBERS OF OBJECTS
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                 Layout   Source        Component Type
                 ------   ------        --------------
 Ports:               7        7

 Nets:               23       23

 Instances:          18       18        MN (4 pins)
                     18       18        MP (4 pins)
                 ------   ------
 Total Inst:         36       36


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Pre-Layout and Post-Layout Parameters for Different PVT conditions.

Fast Fast : 1.32V & -40°

Params	Pre-Layout	Post-Layout
TPLH	79.8 ps	87.4 ps
TPHL	89.3 ps	94.8 ps
TPD	84.557 Ps	91.13 Ps
DYN Power	70.9 µw	77.6 µw

Slow slow : 0.8V & 125°

Params	Pre-Layout	Post-Layout
TPLH	237.3 Ps	258.5 Ps
TPHL	278.8 Ps	290.3 Ps
TPD	258.05 ps	274.4 Ps
DYN Power	17.4 µw	22.8 µw

Typical Typical : 1.0V & 25°

Params	Pre-Layout	Post-Layout
TPLH	119.2 ps	130.3 ps
TPHL	132.2 ps	142.7 ps
TPD	125.7 PS	136.5 PS
DYN Powe	52.2 µw	56.7 µw

Propagation Delay : 258 Ps (Pre Layout) , 274 Ps (Post Layout)

Contamination Delay : 84.5 Ps (Pre Layout) , 91.13 Ps (Post Layout)

Monte Carlo Simulations

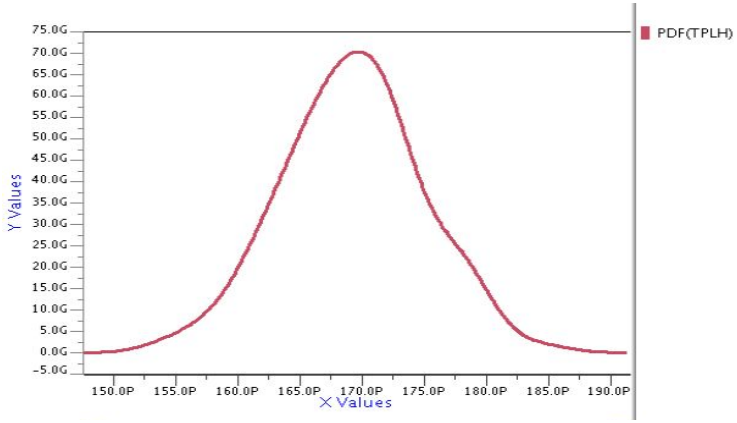

Fig : PDF_TPLH

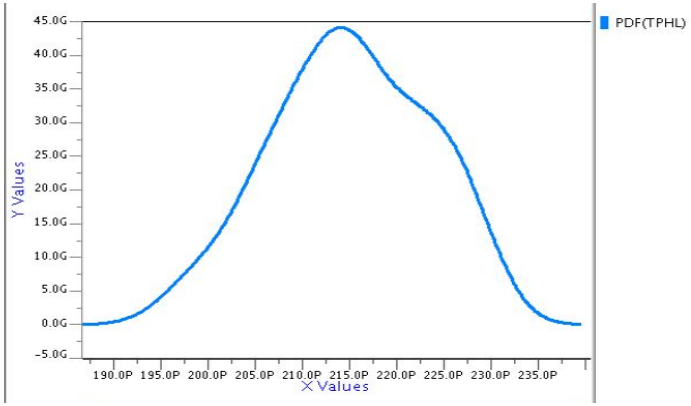

Fig : PDF_TPHL

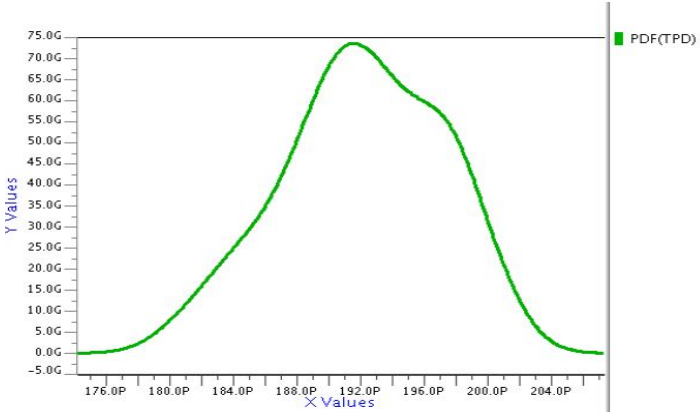

Fig : PDF_Propagation Delay

Work Distribution

Work Distribution:

Debjyoti Musib : Circuit Approach, Simulation Analysis, Sizing, Layout, Pre Layout-Post Layout Simulation, PPT

Gourav Sahu : Xcircuit, Pre Layout Simulation, Nand Layout, Pex, PPT

Sanskar Jain : Circuit Approach, Pre Layout Simulation Connections, Nand Layout.

Aman Mehta : XCircuit Schematic, Connections, PPT